2-Morita Theory of E_2-Algebras and Module-Categories

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Introduction: Morita Equivalence Across Math and Physics

Across mathematics and physics, Morita equivalence asks when different algebraic descriptions encode the same underlying module, boundary, or defect theory.

- **Ring theory.** Classically, two rings are Morita equivalent when their categories of modules are equivalent: Morita 1958.

$$A_1 \overset{\text{1-Morita}}{\simeq} A_2 \quad \Longleftrightarrow \quad \mathcal{C}_{A_1} \simeq \mathcal{C}_{A_2}.$$

- **Rational 2D CFT.** For symmetric special Frobenius algebras A_1, A_2 in a modular tensor category $\mathcal{C}$, Morita equivalence can be described by an invertible bimodule; equivalently, the corresponding module categories are equivalent: Fuchs–Runkel–Schweigert 2002, 2005.

$$A_1 \overset{\text{1-Morita}}{\simeq} A_2 \quad \Longleftrightarrow \quad \exists \text{ an invertible } A_1\text{-}A_2 \text{ bimodule.}$$

Introduction: A Common E_1-Morita Pattern

- **3D TQFTs / $(2+1)$D topological orders.** For fusion categories $\mathcal{A}_1, \mathcal{A}_2$, categorical Morita equivalence is implemented by an invertible bimodule category and is detected by the Drinfeld center: Etingof–Nikshych–Ostrik 2010; Fuchs–Schweigert–Valentino 2013; Carqueville–Runkel 2016; Kong–Wen–Zheng 2015; Kong–Zheng 2022.

$$\mathcal{A}_1 \overset{\text{1-Morita}}{\simeq} \mathcal{A}_2 \quad \Longleftrightarrow \quad \mathfrak{Z}(\mathcal{A}_1) \simeq_{\mathrm{br}} \mathfrak{Z}(\mathcal{A}_2).$$

At the E_1-Morita level, invertible bimodules and equivalences of module categories give the same notion of Morita equivalence. For E_2 and higher, however, establishing the analogous relation requires a higher comparison theorem.

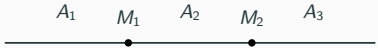
We therefore begin with the common E_1-Morita framework and use it as the template for the higher theory.

E_1 Morita Theory: The E_1-Morita Bicategory

The common E_1-Morita pattern is packaged by a bicategory. Bénabou 1967.

For a finitely cocomplete E_1-monoidal category $\mathcal{C}$ with right-exact tensor product, $\mathbb{M}\mathrm{rt}_{E_1}(\mathcal{C})$ has:

- E_1-algebras $A_i \in \mathcal{A}\mathrm{lg}_{E_1}(\mathcal{C})$ as objects;
- 1-morphisms from $A_1 \to A_2$ are A_1-A_2 bimodules $M_i \in {}_{A_1}\mathcal{C}_{A_2}$;
- 2-morphisms are bimodule homomorphisms;
- 1-morphisms compose via relative tensor product:



$$M_2 \circ M_1 = M_1 \otimes_{A_2} M_2 \in {}_{A_1}\mathcal{C}_{A_3}.$$

Equivalence of objects in $\mathbb{M}\mathrm{rt}_{E_1}(\mathcal{C})$ means that a bimodule has an inverse, i.e.

$$M_1 \otimes_{A_2} M_2 \simeq A_1, \qquad M_2 \otimes_{A_1} M_1 \simeq A_2.$$

E_1 Morita Theory: Equivalence via Invertible Bimodules

Definition (Morita category definition)

Two E_1-algebras are **1-Morita equivalent** if they are equivalent objects in $\mathbb{M}\mathrm{rt}_{E_1}(\mathcal{C})$.

The link with the module-category characterization is given by:

Theorem (Eilenberg–Watts)

For E_1-algebras $A_1, A_2 \in \mathcal{C}$ there is an equivalence Lurie 2017:

$$ {}_{A_1}\mathcal{C}_{A_2} \simeq \mathcal{F}\mathrm{un}_{\mathcal{C}}(\mathcal{C}_{A_1}, \mathcal{C}_{A_2}), \qquad M \longmapsto - \otimes_{A_1} M. $$

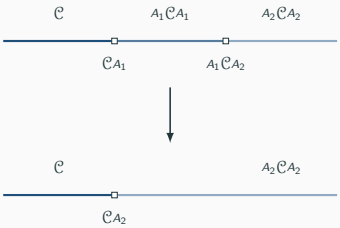
This leads to the module-category realization of the Morita bicategory: $\mathrm{LMod}^{\mathrm{rep}}(\mathcal{C}) \subset \mathrm{LMod}(\mathcal{C})$ of module categories $\mathcal{M}$, module functors, and module natural transformations.

E_1 Morita Theory: Bimodules Are Module Functors

The bicategory $\mathrm{LMod}^{\mathrm{rep}}(\mathcal{C}) \subset \mathrm{LMod}(\mathcal{C})$ consists of

Douglas–Reutter 2018.

- objects are left $\mathcal{C}$-module categories $\mathcal{C}_{A_i}$;
- 1-morphisms from $\mathcal{C}_{A_1}$ to $\mathcal{C}_{A_2}$ are left $\mathcal{C}$-module functor $- \otimes_{A_1} M_i \in \mathcal{F}\mathrm{un}_{\mathcal{C}}(\mathcal{C}_{A_1}, \mathcal{C}_{A_2})$ for $M_i \in {}_{A_1}\mathcal{C}_{A_2}$;
- 2-morphisms are left $\mathcal{C}$-module natural transformations.



Eilenberg–Watts theorem identifies two descriptions of the 1-morphisms:

$$\boxed{\text{algebraic bimodule } M \in {}_{A_1}\mathcal{C}_{A_2}} \quad \longleftrightarrow \quad \boxed{\text{module functor } - \otimes_{A_1} M}.$$

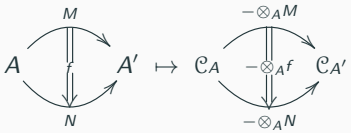
Therefore, an invertible bimodule corresponds precisely to an equivalence of module categories, which recovers the classical characterization of 1-Morita equivalence:

$$A_1 \overset{\text{1-Morita}}{\simeq} A_2 \quad \Longleftrightarrow \quad \mathcal{C}_{A_1} \simeq \mathcal{C}_{A_2}.$$

E_1 Morita Theory: The Module 2-Functor

The Eilenberg–Watts equivalences are compatible with identities $- \otimes_A A \simeq \mathrm{id}_{\mathcal{C}_A}$, composition $(- \otimes_{A'} N) \circ (- \otimes_A M) \simeq - \otimes_A (M \otimes_{A'} N)$ and 2-morphisms $- \otimes_A f$, therefore assemble into the module 2-functor

$$\mathrm{Mod}_1 \colon \mathcal{M}\mathrm{rt}_{E_1}(\mathcal{C}) \longrightarrow \mathrm{LMod}(\mathcal{C}),$$



For each pair A, A', the induced functor on hom-categories is exactly the Eilenberg–Watts equivalence.

Theorem (Module realization)

The 2-functor Mod_1 *is fully faithful.*

The $n = 1$ comparison is now coherent at every morphism level. This becomes the template for $n = 2$.

E_1 Morita Theory: Local Morita Equivalence

Besides the module category $\mathcal{C}_A$, an E_1-algebra A also determines its local endomorphism category

$$\Omega_A(\mathcal{M}\mathrm{rt}_{E_1}(\mathcal{C})) = {}_A\mathcal{C}_A,$$

with monoidal structure given by relative tensor product over A. 1-Morita equivalence can also be characterized by the local module categories under non-degeneracy, i.e. $\mathfrak{Z}_1(\mathcal{C}) \simeq \mathbf{1}$.

Theorem (Local Morita Equivalence)

Let $\mathcal{C}$ be a non-degenerate indecomposable finite multi-tensor category. Then

$$A_1 \overset{1\text{-Morita}}{\simeq} A_2 \iff {}_{A_1}\mathcal{C}_{A_1} \simeq {}_{A_2}\mathcal{C}_{A_2}$$

as E_1-monoidal categories.



In the condensation picture, $\mathcal{C}_A$ is the wall from $\mathcal{C}$ to the condensed phase ${}_A\mathcal{C}_A$. For an indecomposable finite multi-tensor category, these walls compose according to Kong–Zheng 2018.

$$A_1\mathcal{C}A_2\boxed{{}_{A_2}\mathcal{C}_{A_2}}A_2\mathcal{C}A_3 \simeq A_1\mathcal{C}A_3.$$

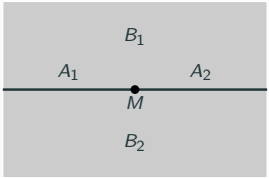
Introduction: E_n-Algebras Encode Strata

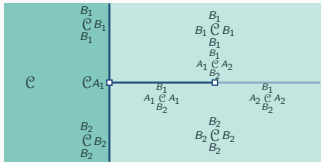
For $n = 1$, full faithfulness of Mod_1 means that passing to module categories loses no bimodule data. This suggests the higher module-realization functor

$$\mathrm{Mod}_n \colon \mathcal{M}\mathrm{rt}_{E_n}(\mathcal{C}) \longrightarrow \mathcal{M}\mathrm{rt}_{E_{n-1}}\big(\mathrm{LMod}(\mathcal{C})\big).$$

E_n-algs $\longrightarrow$ E_{n-1}-alg bimodules $\longrightarrow$ E_{n-2}-alg bimodules between them $\cdots$.

E_{n-1}-module cats $\longrightarrow$ E_{n-2}-bimodule cats $\longrightarrow$ E_{n-3}-bimodule cats between them $\cdots$.





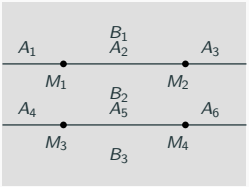
For $n = 2$, the figures above depict the algebraic defect data and its module-category realization. We model E_k-algebras in $\mathcal{C}$ iteratively with $\mathcal{A}\mathrm{lg}_{E_k}(\mathcal{C}) = \mathcal{A}\mathrm{lg}_{E_1}(\mathcal{A}\mathrm{lg}_{E_{k-1}}(\mathcal{C}))$ Lurie 2017, Haugseng 2017.

E_2 Morita Theory: The E_2-Morita tricategory

Following Haugseng's higher-Morita framework Haugseng 2017., we specialize to $n = 2$. For an E_2-monoidal category $\mathcal{C}$, the Morita tricategory $\mathbb{M}\mathrm{rt}_{E_2}(\mathcal{C})$ has:

- objects: E_2-algebras $B_i \in \mathcal{A}\mathrm{lg}_{E_2}(\mathcal{C})$;
- 1-morphisms $B_1 \to B_2$: E_1-algebra B_1-B_2-bimodules

$$A_i \in {}_{B_1}\mathcal{A}\mathrm{lg}_{E_1}(\mathcal{C})_{B_2} \simeq \mathcal{A}\mathrm{lg}_{E_1}\left({}_{B_1}\mathcal{C}_{B_2}\right);$$



- 2-morphisms $A_1 \Rightarrow A_2$: E_0 B_1-B_2-bi-A_1-A_2-modules

$$M_i \in {}_{A_1}{}^{B_1}_{B_2}\mathcal{C}_{A_2} \simeq \mathcal{A}\mathrm{lg}_{E_0}\left({}_{A_1}\left({}_{B_1}\mathcal{C}_{B_2}\right)_{A_2}\right);$$

- 3-morphisms: compatible bimodule homomorphisms.

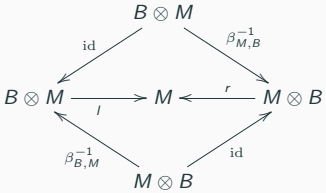
Compositions of 1- and 2-morphisms are given by the appropriate relative tensor products.

E_2 Morita Theory: Local Modules as Bi-bimodules

For the identity 1-morphism $B : B \to B$, we can take $A_1 = A_2 = B$. Then a 2-endomorphism is a bi-bimodule $M \in {}_B{}^B_B\mathcal{C}_B$, and its coherence condition is the commutative diamond.

This is the usual locality condition for a B-module. Fuchs et al. 2006; Kong 2014.

Moreover, the two compositions of these 2-endomorphisms induce the canonical E_2-monoidal structure, corresponding to the relative tensor product and braiding of local modules.



Theorem

There is an E_2-monoidal equivalence ${}_B{}^B_B\mathcal{C}_B \simeq \mathcal{C}_B{}^{\mathrm{loc}}$.

E_2 Morita Theory: 2-Morita Equivalence

Definition

Two E_2-algebras are **2-Morita equivalent** if they are equivalent objects in $\mathbb{M}\mathrm{rt}_{E_2}(\mathcal{C})$.

Thus an equivalence is given by inverse 1-morphisms $A : B_1 \to B_2$ and $A' : B_2 \to B_1$:

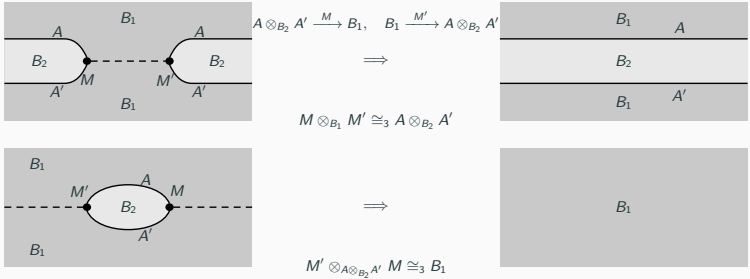


Figure 1: Here we display the B_1 side of the inverse data. The 2-morphisms M and M' are mutually inverse: the upper row reduces their composite to the identity 2-morphism on $A \otimes_{B_2} A'$, while the lower row reduces the reverse composite to the identity 2-morphism on B_1.

E_2 Morita Theory: E_2-Algebras Yield Monoidal Modules

Now we realize the three functor

$$\mathrm{Mod}_2 \colon \mathcal{M}\mathrm{rt}_{E_2}(\mathcal{C}) \longrightarrow \mathcal{M}\mathrm{rt}_{E_1}\big(\mathrm{LMod}(\mathcal{C})\big).$$

explicitly.

The module realization begins with

$$B \in \mathcal{A}\mathrm{lg}_{E_2}(\mathcal{C}) \quad \longmapsto \quad \mathcal{C}_B.$$

Because B is an E_2-algebra, $\mathcal{C}_B$ is naturally an E_1-monoidal left $\mathcal{C}$-module category, with tensor product given by $\otimes_B$.

Let

$$A \in {}_{B_1}\mathcal{A}\mathrm{lg}_{E_1}(\mathcal{C})_{B_2} = \mathcal{A}\mathrm{lg}_{E_1}\begin{pmatrix} B_1 \\ \mathcal{C} \\ B_2 \end{pmatrix}.$$

As A is an E_1-algebra in the monoidal category $\begin{smallmatrix} B_1 \\ \mathcal{C} \\ B_2 \end{smallmatrix}$. Applying the E_1 module construction internally gives

$$A \quad \longmapsto \quad \begin{smallmatrix} B_1 \\ \mathcal{C}_A \\ B_2 \end{smallmatrix}$$

E_2 Morita Theory: 1-Morphisms Become Bimodule Categories

In fact, ${}^{B_1}_{B_2}\mathcal{C}_A$ can be dentified with the familiar category $\mathcal{C}_A$:

Theorem

The forgetful functor

$$F : {}_{B_i\otimes B_j}{}^{B_i}_{B_j}\mathcal{C}_A \simeq {}^{B_i}_{B_j}\mathcal{C}_A \longrightarrow \mathcal{C}_A \simeq {}_1{}^{1}_{1}\mathcal{C}_A,$$

$$(M, \lambda_{ij}, \rho_A) \longmapsto (M, \rho_A)$$

is an equivalence of categories.

For fixed endpoint data, the compatible outer actions carry no additional information: they are canonically determined by the A-action. Consequently,

$$A : B_1 \to B_2 \qquad \longmapsto \qquad \mathcal{C}_A : \mathcal{C}_{B_1} \to \mathcal{C}_{B_2},$$

where $\mathcal{C}_A$ carries the corresponding $\mathcal{C}_{B_1}$-$\mathcal{C}_{B_2}$ bimodule-category structure.

E_2 Morita Theory: 2-Morphisms Become Bimodule Functors

Let B_1, B_2 be E_2-algebras and $A_1, A_2 \in {}_{B_1}\mathcal{A}\mathrm{lg}_{E_1}(\mathcal{C})_{B_2}$.

Theorem (E_2 Eilenberg–Watts)

There is an equivalence

$$\begin{matrix} B_1 \\ A_1 \mathcal{C} A_2 \\ B_2 \end{matrix} \simeq \begin{matrix} \mathcal{C}_{B_1} \\ \mathfrak{C}\mathcal{F}\mathrm{un}(\mathcal{C}A_1, \mathcal{C}A_2). \\ \mathcal{C}_{B_2} \end{matrix}$$

Under this equivalence, 2-morphisms are sent to functors of the form

$$M \longmapsto - \otimes_{A_1} M : \mathcal{C}A_1 \longrightarrow \mathcal{C}A_2.$$

Thus a 2-morphism $M : A_1 \Rightarrow A_2$ in $\mathbb{M}\mathrm{rt}_{E_2}(\mathcal{C})$ is realized as a $\mathcal{C}_{B_1}$-$\mathcal{C}_{B_2}$ bimodule functor in $\mathrm{LMod}(\mathcal{C})$.

Likewise, a 3-morphism $f : M_1 \to M_2$ is sent to the natural transformation in $\mathrm{LMod}(\mathcal{C})$

$$f : M_1 \to M_2 \qquad \longmapsto \qquad - \otimes_{A_1} f.$$

Theorem (Module 3-functor)

The 3-functor Mod_2 *is 2-fully faithful.*

E_2 Morita Theory: The Representable Image

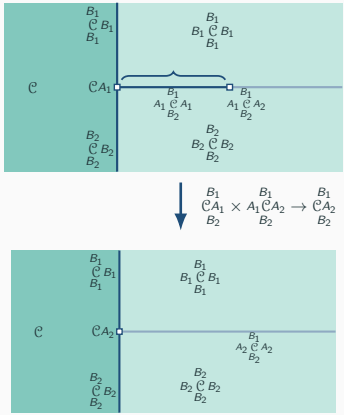
Mod_2 is a triequivalence onto its representable essential image;

$$\mathcal{M}\mathrm{rt}_{E_2}(\mathcal{C}) \xrightarrow[\simeq]{\mathrm{Mod}_2} \mathcal{M}\mathrm{rt}_{E_1}\left(\mathrm{LMod}^{\mathrm{rep}}(\mathcal{C})\right) \hookrightarrow \mathcal{M}\mathrm{rt}_{E_1}\left(\mathrm{LMod}(\mathcal{C})\right).$$

nonrepresentable bimodule categories may still exist in the full target. We therefore isolate $\mathcal{M}\mathrm{rt}_{E_1}\left(\mathrm{LMod}^{\mathrm{rep}}(\mathcal{C})\right)$ as:

- objects are left $\mathcal{C}$-module categories $\mathcal{C}_{B_i}$;
- 1-morphisms from $\mathcal{C}_{B_1}$ to $\mathcal{C}_{B_2}$ are left $\mathcal{C}$-module categories ${}^{B_1}_{B_2}\mathcal{C}_{A_i}$;
- 2-morphisms from ${}^{B_1}_{B_2}\mathcal{C}_{A_1}$ to ${}^{B_1}_{B_2}\mathcal{C}_{A_2}$ are left $\mathcal{C}$-module functors $-\otimes_{A_1} M_i \in {}_{\mathcal{C}B_2}{}^{\mathcal{C}B_1}\mathcal{F}\mathrm{un}(\mathcal{C}_{A_1}, \mathcal{C}_{A_2})$ for $M_i \in {}_{A_1}{}^{B_1}_{B_2}\mathcal{C}_{A_2}$;
- 3-morphisms are left $\mathcal{C}$-module natural transformations.

E_2 Eilenberg–Watts 2-morphism realized as $-\otimes_{A_1} M$



E_2 Morita Theory: Local Data Detects 2-Morita Equivalence

For an E_2-algebra B, its double loop recovers the local module category:

$$\Omega_B^2\big(\mathcal{M}\mathrm{rt}_{E_2}(\mathcal{C})\big) = {}_B{}^{B}_{B}\mathcal{C}_B \simeq \mathcal{C}_B{}^{\mathrm{loc}}.$$

In the modular fusion setting, this twice-iterated endomorphism data is a complete invariant of the 2-Morita class:

Theorem (Local-to-global criterion)

Let $\mathcal{C}$ be a modular fusion category. For condensable E_2-algebras $B_1, B_2 \in \mathcal{C}$,

$$B_1 \overset{2\text{-}Morita}{\simeq} B_2 \quad \Longleftrightarrow \quad {}_{B_1}{}^{B_1}_{B_1}\mathcal{C}_{B_1} \simeq_{E_2} {}_{B_2}{}^{B_2}_{B_2}\mathcal{C}_{B_2}.$$

Xu–Yang 2025, Prop. C.11.

Thus, in the non-degenerate setting relevant to topological orders, the local defect data determines the global 2-Morita class.

E_2 Morita Theory: Fusion under Module Realization

Beyond comparing defects, the Morita category also encodes their fusion. Let $\mathcal{C}$ be an indecomposable finite braided multi-tensor category:

Horizontal fusion

Fixed E_2-phases B_1, B_2

For $A_1, A_2, A_3 : B_1 \longrightarrow B_2$, we have

$$\begin{matrix} & B_1 & \\ A_1 & \mathcal{C} & A_2 \\ & B_2 & \end{matrix} \boxed{\begin{matrix} B_1 \\ A_1 \mathcal{C} A_2 \\ B_2 \end{matrix}} \begin{matrix} & B_1 & \\ A_2 & \mathcal{C} & A_3 \\ & B_2 & \end{matrix} \simeq \begin{matrix} & B_1 & \\ A_1 & \mathcal{C} & A_3. \\ & B_2 & \end{matrix}$$

Under the iterated-bimodule equivalence, This is ordinary composition in

$$\mathfrak{Mrt}_{E_1}({}_{B_1}\mathcal{C}_{B_2}).$$

Vertical fusion

Fuse over the intermediate E_2-phase B_2

For $A_1, A_2 : B_1 \to B_2$ and $A_3, A_4 : B_2 \to B_3$,

$$\begin{matrix} & B_1 & \\ A_1 & \mathcal{C} & A_2 \\ & B_2 & \end{matrix} \boxed{\begin{matrix} B_2 \\ A_1 B_1 A_2 \\ B_2 \end{matrix}} \begin{matrix} & B_2 & \\ A_3 & \mathcal{C} & A_4 \\ & B_3 & \end{matrix} \simeq \begin{matrix} & A_1 B_1 A_2 & \\ \textcircled{B_2} & \mathcal{C} & \textcircled{B_2}. \\ & A_3 B_3 A_4 & \end{matrix}$$

The intermediate B_2-phase is removed, while the adjacent E_1-algebras compose over B_2:

$$A_1 \otimes_{B_2} A_3, \qquad A_2 \otimes_{B_2} A_4.$$

Toric Code: A Concrete Fusion Rule

Connected condensable E_2-algebras in the Toric Code

$$\mathcal{TC} = \mathfrak{Z}(\mathrm{Vec}_{\mathbb{Z}_2}),$$

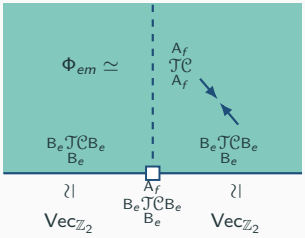
$$\begin{aligned} \mathrm{B}_0 &= 1 && \text{no condensation,} \\ \mathrm{B}_e &= 1 \oplus e && \text{electric condensation,} \\ \mathrm{B}_m &= 1 \oplus m && \text{magnetic condensation.} \end{aligned}$$

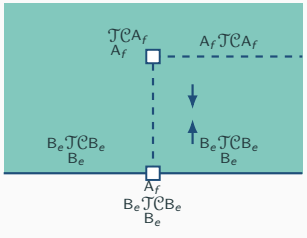
The connected separable skeleton has 18 canonical 1-morphisms. For this fusion rule we only need

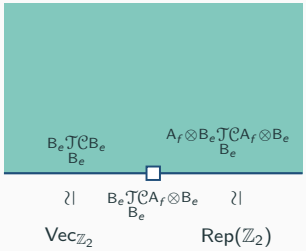
$$\mathrm{A}_f : \mathrm{B}_0 \to \mathrm{B}_0, \qquad \Phi_{em} \simeq \underset{\mathrm{A}_f}{\overset{\mathrm{A}_f}{\mathcal{TC}}},$$

Fusing Φ_{em} with the rough boundary gives the smooth boundary:

$$\underset{\mathrm{B}_e}{\overset{\mathrm{A}_f\otimes\mathrm{B}_e}{\mathcal{TC}}}{}_{\mathrm{A}_f\otimes\mathrm{B}_e} \simeq \underset{\mathrm{B}_e}{\overset{\mathrm{B}_m\otimes\mathrm{B}_e}{\mathcal{TC}}}{}_{\mathrm{B}_m\otimes\mathrm{B}_e} \simeq \underset{\mathrm{B}_m}{\overset{\mathrm{B}_m}{\mathcal{TC}}}{}_{\mathrm{B}_m}$$

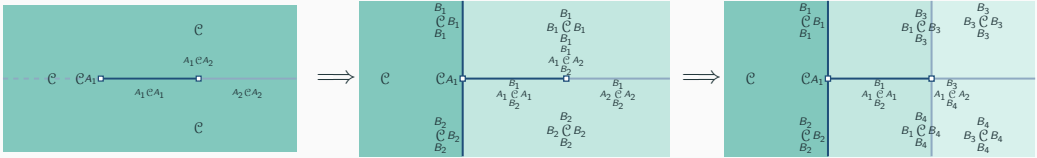






Iterating the Opening: Two-Step Condensation

The first condensation is governed by the unit maps $1 \hookrightarrow B_i$: forgetting the transverse B_1–B_2 actions recovers the ordinary A_1–A_2 bimodule in $\mathcal{C}$, while choosing compatible outer actions resolves it as a bi-bimodule.



The same mechanism can be iterated. For

$$f_1 : B_1 \hookrightarrow B_3, \qquad f_2 : B_2 \hookrightarrow B_4,$$

if the B_1–B_2 actions on A_1, A_2 and M extend along (f_1, f_2), restriction of scalars gives

$$(f_1, f_2)^* : {}_{A_1}\overset{B_3}{\underset{B_4}{\mathcal{C}}}{}_{A_2} \hookrightarrow {}_{A_1}\overset{B_1}{\underset{B_2}{\mathcal{C}}}{}_{A_2} \longrightarrow {}_{A_1}\mathcal{C}_{A_2}.$$

Thus two-step condensation is obtained by successively opening the outer E_2-actions.

Automorphism-Induced Walls and Closing

Let B be a condensable E_2-algebra. An algebra automorphism $\varphi \in \mathrm{Aut}_{\mathcal{A}\mathrm{lg}_{E_2}(\mathcal{C})}(B)$ can twist one of the regular B-actions.
In the intermediate B-condensed phase, this gives an invertible wall

$$\Phi_\varphi \quad \leftrightsquigarrow \quad {}_{B^{\mathrm{tw}}}\overset{B}{\mathcal{C}}_{B^{\mathrm{tw}}},$$

which can be genuinely non-trivial.

Since

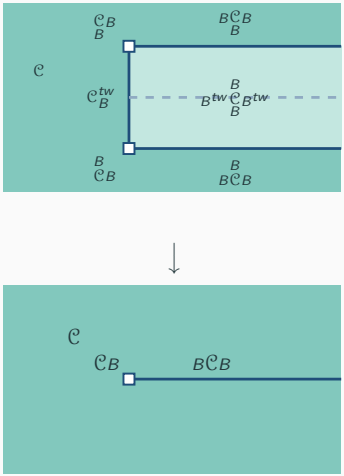
$$\begin{matrix} B \\ \circledB \\ B^{\mathrm{tw}} \\ \circledB \\ B \end{matrix} \simeq \begin{matrix} 1 \\ B^{\mathrm{tw}} \\ 1 \end{matrix} \cong B.$$

Both the wall and its endpoint reduce to the untwisted ones when Φ_φ is with the two B-condensation sides:

$${}_{B^{\mathrm{tw}}}\overset{B}{\mathcal{C}}_{B^{\mathrm{tw}}} \xrightarrow{\text{vertical closing}} {}_B\mathcal{C}_B, \qquad \mathcal{C}_{B^{\mathrm{tw}}} \longrightarrow \mathcal{C}_B.$$

Theorem

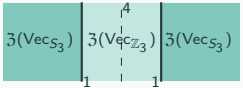
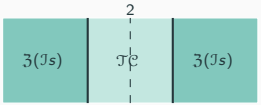
An invertible domain wall induced by an algebra automorphism does not change the resulting defect after closing with the condensation walls.



Two Examples of Automorphism-Induced Walls

- In $\mathfrak{Z}(\mathrm{Vec}_{S_3})^{\mathrm{loc}}_{A\oplus B} \simeq \mathfrak{Z}(\mathrm{Vec}_{\mathbb{Z}_3})$, a non-trivial automorphism of $A \oplus B$ induces the braided autoequivalence ϕ_{1-2}.
- In $\mathfrak{Z}(\mathcal{I}s)^{\mathrm{loc}}_{A_2} \simeq \mathcal{TC}$, the non-trivial automorphism $(1 \boxtimes \overline{1}) \oplus (\psi \boxtimes \overline{\psi}) \xrightarrow{1\oplus(-1)} (1 \boxtimes \overline{1}) \oplus (\psi \boxtimes \overline{\psi})$ induces the **e–m** exchange.

By the previous proposition, walls differing only by such an automorphism-induced wall become identified after closing. Hence the intermediate walls are grouped into automorphism orbits.

E_2 Condensable algebras	Condensed phase	Walls before closing	# Walls after closing
$A \oplus B$ in $\mathfrak{Z}(\mathrm{Vec}_{S_3})$	$\mathfrak{Z}(\mathrm{Vec}_{\mathbb{Z}_3})$		$4/2 = 2$
$(1 \boxtimes \overline{1}) \oplus (\psi \boxtimes \overline{\psi})$ in $\mathfrak{Z}(\mathcal{I}s)$	$\mathcal{TC}$		$2/2 = 1$

E_n Morita Theory: A Recursive Pattern

The $n = 1$ and $n = 2$ cases are the first layers of a recursive construction. For E_n-algebras $X_1, X_2 \in \mathcal{C}$, the mapping n-category is itself a lower Morita category Haugseng 2017.:

$$\mathrm{Hom}_{\mathcal{M}\mathrm{rt}_{E_n}(\mathcal{C})}(X_1, X_2) \simeq \mathcal{M}\mathrm{rt}_{E_{n-1}}\big(X_1 \mathcal{C} X_2\big).$$

Thus an E_n-Morita problem between X_1 and X_2 becomes an E_{n-1}-Morita problem inside their bimodule category.

Proposition

The two natural descriptions of a 1*-morphism agree:*

$$X_1 \mathcal{A}\mathrm{lg}_{E_{n-1}}(\mathcal{C}) X_2 \simeq \mathcal{A}\mathrm{lg}_{E_{n-1}}\big(X_1 \mathcal{C} X_2\big).$$

The same recursive pattern appears after passing to module categories:

$$\mathrm{Mod}_n : \mathcal{M}\mathrm{rt}_{E_n}(\mathcal{C}) \longrightarrow \mathcal{M}\mathrm{rt}_{E_{n-1}}\big(\mathrm{LMod}(\mathcal{C})\big), \qquad X \longmapsto \mathcal{C} x.$$

For $n = 1, 2$, Mod_n is n-fully faithful; we expect the same pattern in higher dimensions.